

iSHAP: Rule-Guided Synthetic Data Augmentation for Better Feasibility–Performance Trade-offs

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Abstract

We present a practical workflow for steering model behavior with lightweight structural rules while preserving predictive performance. The key idea is to iteratively generate synthetic training points and optimize them against a combined objective: held-out model performance (RMSE) and a SHAP-derived feasibility score that measures alignment with user-defined directional rules. Unlike post-hoc explanation-only methods, our approach actively edits training data to improve causal plausibility of learned effects. We provide an implementation based on LightGBM, Optuna, and TreeSHAP, and demonstrate the pipeline on two tabular datasets.

1 Introduction

Many machine learning systems achieve strong predictive metrics but violate simple domain expectations. For example, in glucose forecasting one may expect that increased insulin should decrease future glucose, yet unconstrained models may learn the opposite or unstable directions. This creates a gap between pure predictive optimization and rule-consistent behavior.

We propose iSHAP, a method that uses structural rules and SHAP-based diagnostics to guide synthetic augmentation. At each round, candidate synthetic points are optimized with respect to a combined objective balancing:

1. prediction quality on hold-out data, and
2. rule feasibility measured from SHAP directional correlations.

If a candidate improves the combined objective, it is added to training data and the model is retrained. This creates a closed loop between model fit, explanation diagnostics, and data augmentation.

2 Method

2.1 Problem Setup

Given training data (X, y) and a set of directed rules $\mathcal{R} = \{(u \rightarrow v, s)\}$ with desired sign $s \in \{-1, 0, +1\}$, we train a base model and iteratively add synthetic points. For each rule, feasibility is computed by correlating SHAP effects with corresponding features. For target rules $(u \rightarrow y)$, we use

$$\rho_u = \text{corr}(\phi_u(x), x_u),$$

and compare ρ_u to target sign s . Aggregating weighted deviations across rules yields a feasibility score in $[0, 1]$.

2.2 Optimization Objective

At each round, Optuna (TPE sampler) proposes synthetic points. Each trial is scored by a combined objective that rewards lower RMSE and higher feasibility. The trade-off is controlled by `objective_rmse_weight`:

- 0.0: prioritize RMSE only,
- 1.0: prioritize feasibility only,
- intermediate values: linear trade-off.

2.3 Round-Based Augmentation

The pipeline:

1. Train baseline model on current augmented set.
2. Optimize a batch of synthetic points.
3. Evaluate combined score on fixed validation diagnostics.
4. Accept batch if combined score improves; otherwise stop.

3 Experimental Setup

3.1 Datasets

We currently evaluate:

- **d1namo**: glucose-related tabular regression task.
- **cancer_air_pollution**: cancer-level dataset with air-pollution risk factors.

3.2 Rules

For controlled sanity checks, we use one rule per dataset:

- **d1namo**: `insulin` $\rightarrow$ `future glucose`, negative.
- **cancer_air_pollution**: `Air Pollution` $\rightarrow$ `Level`, positive.

4 Results

4.1 Quantitative Summary

Table 1 summarizes the current runs. For both datasets, the approach improves feasibility and hold-out RMSE, indicating that rule alignment can be increased without sacrificing predictive quality in these initial experiments.

For **d1namo**, rule alignment improved substantially: `insulin` $\rightarrow$ `future glucose` moved from Pearson -0.4299 to -0.6289 , closer to the desired negative direction. At the same time, RMSE improved by -0.3074 . For **cancer_air_pollution**, the directional rule `Air Pollution` $\rightarrow$ `Level` improved from 0.6612 to 0.6661 , while RMSE improved from 0.0066 to 0.0023 .

Table 1: Before/after results from current pipeline runs.

Dataset	Δ Feasibility	RMSE Before	RMSE After	Synth Rows Added
d1namo	+0.0995	28.3282	28.0208	30
cancer_air_pollution	+0.0025	0.0066	0.0023	10

4.2 Visual Evidence

Figures 1–4 show before/after explanation shifts. In d1namo, beeswarm and interaction patterns indicate a stronger directional consistency with the insulin rule after augmentation. In cancer_air_pollution, the visuals show a smaller but consistent directional tightening around the selected rule.

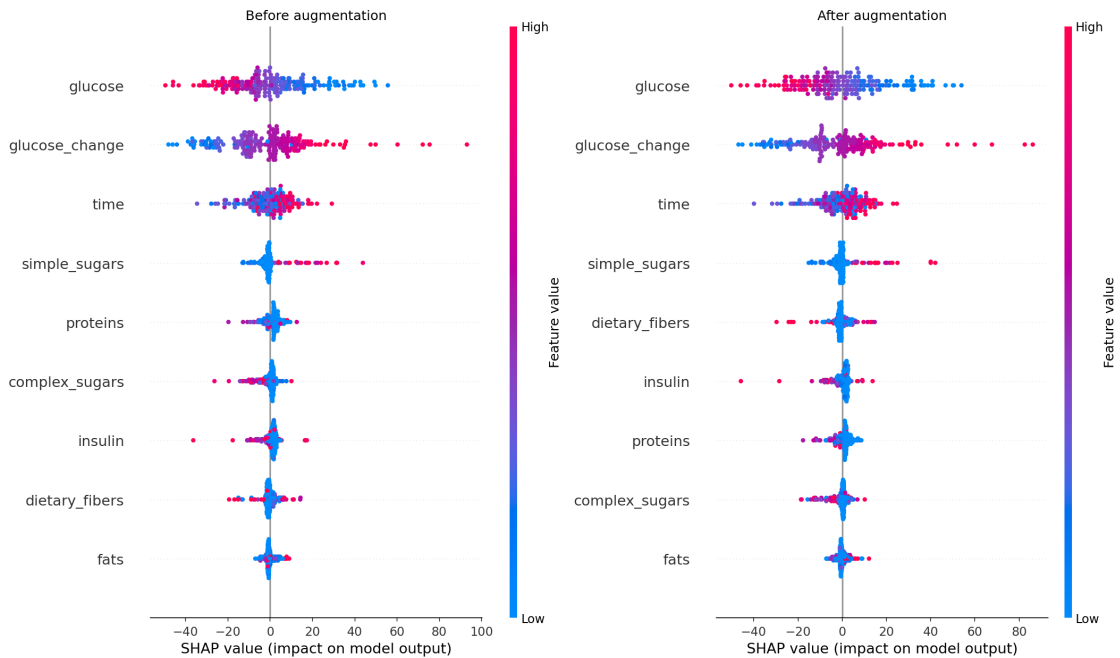


Figure 1: d1namo SHAP beeswarm before and after augmentation.

5 Discussion

These initial results support the central premise of iSHAP: synthetic augmentation can move a model toward predefined directional rules while preserving or improving predictive performance. The gains are strongest in d1namo, where the rule conflict was larger at baseline and thus allowed more directional correction. The cancer dataset shows a smaller feasibility gain, likely because baseline alignment was already relatively high.

A practical benefit is interpretability of progress at each round: the feasibility metric, per-rule Pearson shifts, and explanation plots all move in a coherent direction. This makes the optimization process auditable, rather than relying only on aggregate predictive metrics.

Overall, the current evidence suggests that rule-guided augmentation is a viable mechanism to reduce explanation–domain mismatch in tabular settings. Further validation across additional

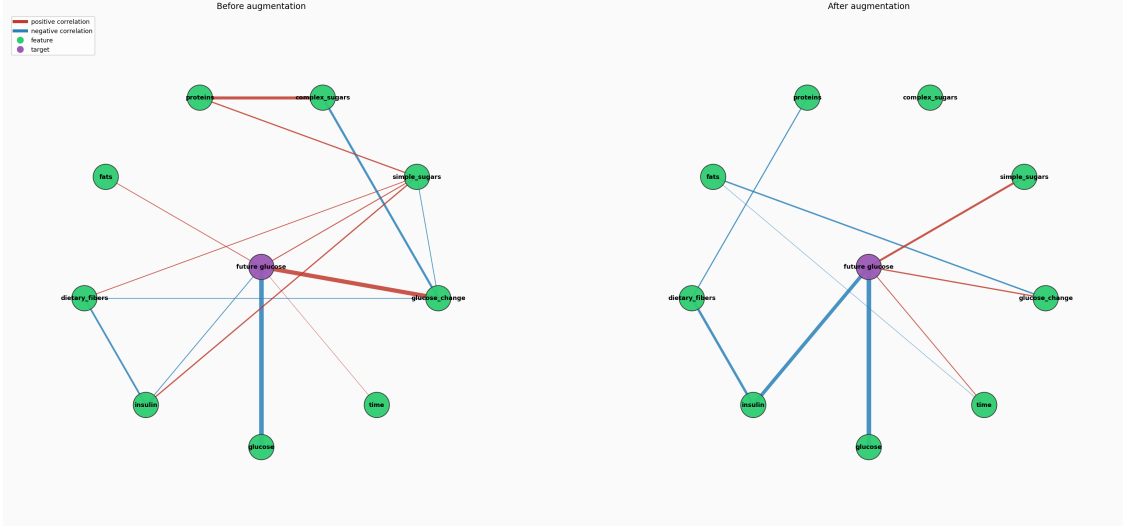


Figure 2: dlnamo correlation-based interaction network before and after augmentation.

datasets and richer SCM rule sets is needed to establish robustness.

6 Limitations and Future Work

Current limitations include potential search inefficiency in high-dimensional spaces and dependence on SHAP stability. Future work should investigate stronger structural constraints, alternative attribution methods, and more robust optimization schemes.

7 Conclusion

iSHAP offers a practical framework for injecting domain rules into model development through data-centric optimization. By coupling synthetic augmentation with explanation-based feasibility scoring, practitioners can iteratively move models toward both stronger performance and more plausible directional behavior.

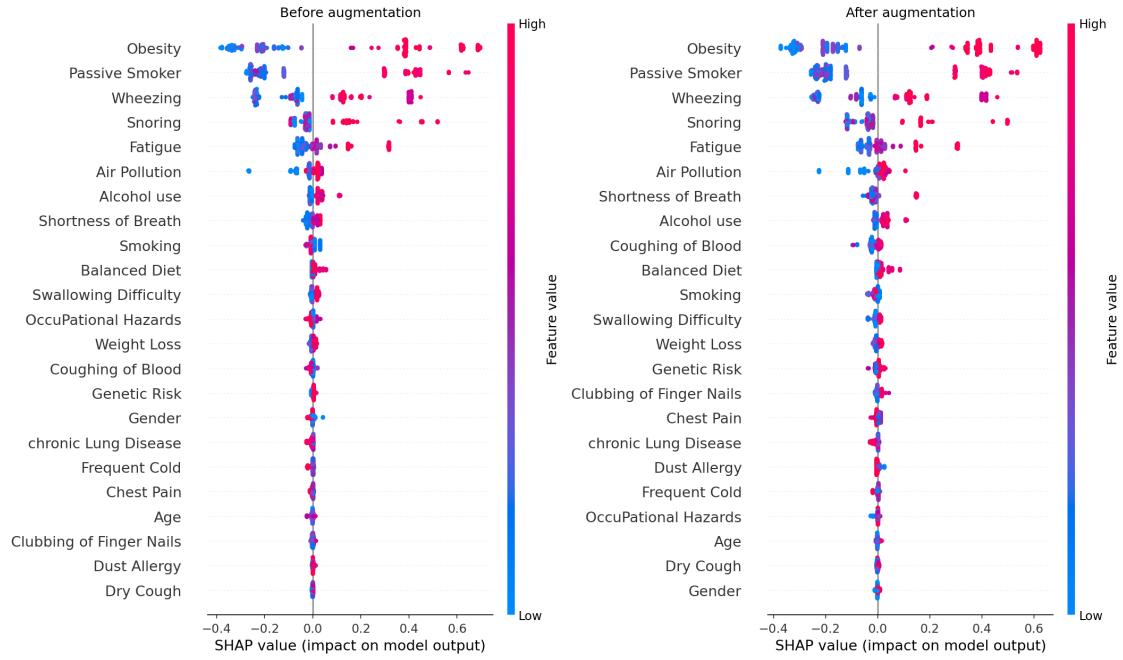


Figure 3: cancer_air_pollution SHAP beeswarm before and after augmentation.

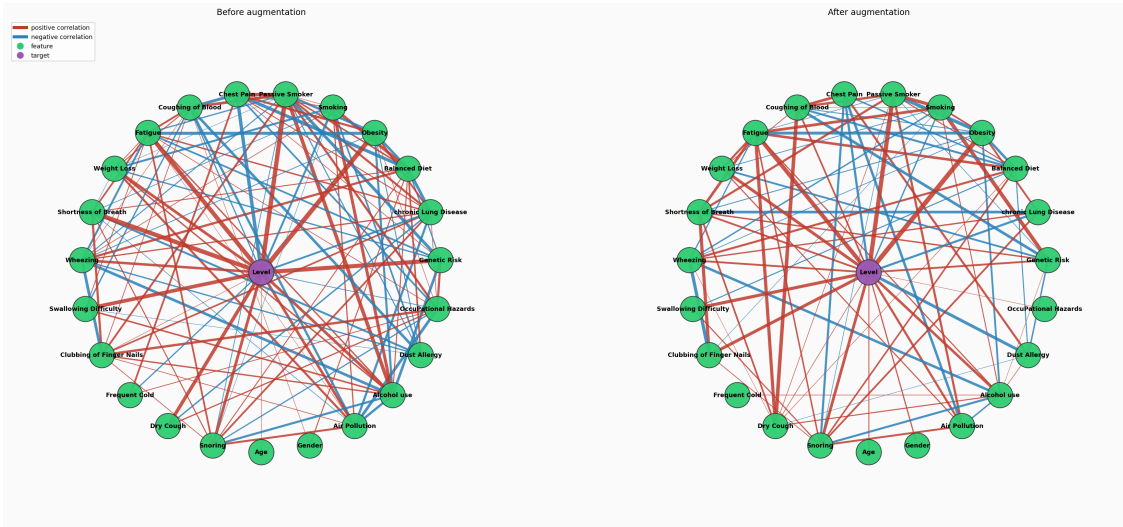


Figure 4: cancer_air_pollution correlation-based interaction network before and after augmentation.